P-201 Crude Oil Transfer Pump · Operation & Maintenance Manual

BPCL Mathura Refinery · Rotating Equipment Division

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1. Equipment Description

Asset Tag: P-201. Type: Horizontal Single-Stage Centrifugal Pump. Service: Crude Oil Transfer from Tank Farm TK-101 to CDU Feed. Manufacturer: Flowserve Corporation. Model: PVXM 200. Year: 2018. Plant: BPCL Mathura Refinery, Unit 2 · Crude Distillation Unit (CDU).

2. Design Parameters

Rated Flow: 850 m3/hr. Rated Head: 145 m. Speed: 1480 RPM. Motor Power: 450 kW. Operating Temperature: 40 degC. Suction Pressure: 0.3 bar g. Discharge Pressure: 15.2 bar g. Specific Gravity: 0.85. Viscosity: 12 cSt. NPSH Required: 4.2 m.

3. Bearing Specifications

Drive End Bearing: SKF 6316 (Deep Groove Ball Bearing). Non-Drive End Bearing: SKF NU316 (Cylindrical Roller Bearing). Bearing Inner Race Fault Frequency (BPFI): 103.4 Hz at 1480 RPM. Bearing Outer Race Fault Frequency (BPFO): 68.2 Hz at 1480 RPM. Ball Pass Frequency (BSF): 45.1 Hz. Cage Frequency: 8.9 Hz. Lubrication: Mobil SHC 100 grease. Regreasing interval: 1000 hours.

4. Vibration Acceptance Criteria

Per ISO 10816-3 and API 610 12th Edition: Zone A (New machine acceptable): 0 to 2.8 mm/s RMS. Zone B (Long-term operation acceptable): 2.8 to 7.1 mm/s RMS. Zone C (Alarm threshold · restricted operation): 7.1 to 11.2 mm/s RMS. Zone D (Danger · immediate shutdown required): greater than 11.2 mm/s RMS. High vibration alarm setpoint: 7.1 mm/s. Shutdown setpoint: 11.2 mm/s.

5. Mechanical Seal

Type: John Crane Type 2800 Double Mechanical Seal. Seal Flush Plan: API Plan 53B (pressurized barrier fluid). Barrier Fluid Pressure: 1.5 bar above seal chamber pressure. Barrier Fluid: Shell Tellus S2 M 46 (hydraulic oil). Seal flush flow rate: 2·3 LPM. Maximum seal temperature: 80 degC.

6. Bearing Replacement Interval

Replace bearings every 8,000 operating hours OR when vibration exceeds 7.1 mm/s RMS (Zone C threshold), whichever occurs first. Condition-based replacement recommended if bearing temperature exceeds 85 degC or if envelope spectrum shows BPFI/BPFO peaks greater than 3x running speed amplitude.

7. LOTO (Lockout-Tagout) Procedure

STEP 1: Notify Control Room operator and obtain Permit to Work (PTW). STEP 2: Close suction isolation valve V-201A (handwheel, 5 turns CW). STEP 3: Close discharge isolation valve V-201B (handwheel, 8 turns CW). STEP 4: De-energize MCC-2 breaker CB-201 (Lock with personal padlock). STEP 5: Tag out at both valve handwheels and MCC breaker. STEP 6: Open vent valve V-201C to depressurize casing. STEP 7: Verify zero energy state with calibrated pressure gauge.

8. Maintenance Schedule

Daily: Check bearing temperature (max 85 degC), seal flush pressure (1.5 bar), vibration level (max 7.1 mm/s). Monthly: Grease bearing housings (15g each).

Quarterly: Check coupling alignment (max 0.05 mm offset, 0.03 mm angular). Annually: Inspect mechanical seal, replace O-rings and wear rings. 8000 hours: Complete bearing replacement, clearance check, impeller inspection.

9. Common Fault Signatures

CAVITATION: Crackling noise, high vibration at vane pass frequency (VPF = 5x RPM). Check NPSH margin. Throttle discharge slightly. BEARING WEAR: High vibration at BPFI (103 Hz) or BPFO (68 Hz). Plan bearing change. MISALIGNMENT: High vibration at 2x RPM (49.3 Hz). Check coupling. IMBALANCE: High vibration at 1x RPM (24.7 Hz). Balance impeller. SEAL FAILURE: Barrier fluid level drop more than 10% in 24 hours. Emergency shutdown.

10. Spare Parts List

SKF 6316 Bearing (Drive End): BPCL SAP Material No. 10045612. SKF NU316 Bearing (NDE): BPCL SAP Material No. 10045613. John Crane 2800 Seal Kit: BPCL SAP Material No. 20087234. Coupling Insert (Falk Steelflex T10): BPCL SAP Material No. 30012456. Impeller Wear Ring Set: BPCL SAP Material No. 10098765. Minimum stock: 2 complete bearing sets always on hand (critical rotating equipment).